

```
In [1]: import os
os.getcwd()
```

```
Out[1]: 'C:\\Users\\kalki\\climate-data-challenge-week0\\notebooks'
```

```
In [2]: import os
os.listdir("../data")
```

```
Out[2]: ['compare_countries.csv',
'compare_countries_clean.csv',
'ethiopia.csv',
'ethiopia_clean.csv',
'kenya.csv',
'kenya_clean.csv',
'nigeria.csv',
'nigeria_clean.csv',
'sample.csv',
'sudan.csv',
'sudan_clean.csv',
'tanzania.csv',
'tanzania_clean.csv']
```

```
In [3]: import pandas as pd

ethiopia = pd.read_csv("../data/ethiopia_clean.csv")
kenya = pd.read_csv("../data/kenya_clean.csv")
nigeria = pd.read_csv("../data/nigeria_clean.csv")
sudan = pd.read_csv("../data/sudan_clean.csv")
tanzania = pd.read_csv("../data/tanzania_clean.csv")
```

```
In [4]: ethiopia["country"] = "Ethiopia"
kenya["country"] = "Kenya"
nigeria["country"] = "Nigeria"
sudan["country"] = "Sudan"
tanzania["country"] = "Tanzania"
```

```
In [5]: df = pd.concat([ethiopia, kenya, nigeria, sudan, tanzania])

df.head()
```

Out[5]:

	YEAR	DOY	T2M	T2M_MAX	T2M_MIN	T2M_RANGE	PRECTOTCORR	RH2M	WS2M
0	2015	1	25.18	35.55	16.14	19.41	0.0	31.87	3.46
1	2015	2	23.92	34.14	15.81	18.33	0.0	23.83	4.24
2	2015	3	22.73	31.64	15.09	16.55	0.0	38.21	5.01
3	2015	4	19.15	27.35	12.88	14.47	0.0	21.07	5.72
4	2015	5	17.54	27.22	9.49	17.73	0.0	21.58	4.28



In [6]: `df["date"] = pd.to_datetime(df["date"])
df["year"] = df["date"].dt.year
df["month"] = df["date"].dt.month`

In [7]: `df["country"].value_counts()`

Out[7]:

```
country
Ethiopia    4108
Kenya       4108
Nigeria     4108
Sudan       4108
Tanzania    4108
Name: count, dtype: int64
```

In [8]: `monthly_temp = df.groupby(["country", "year", "month"])["T2M"].mean().reset_index()

monthly_temp.head()`

Out[8]:

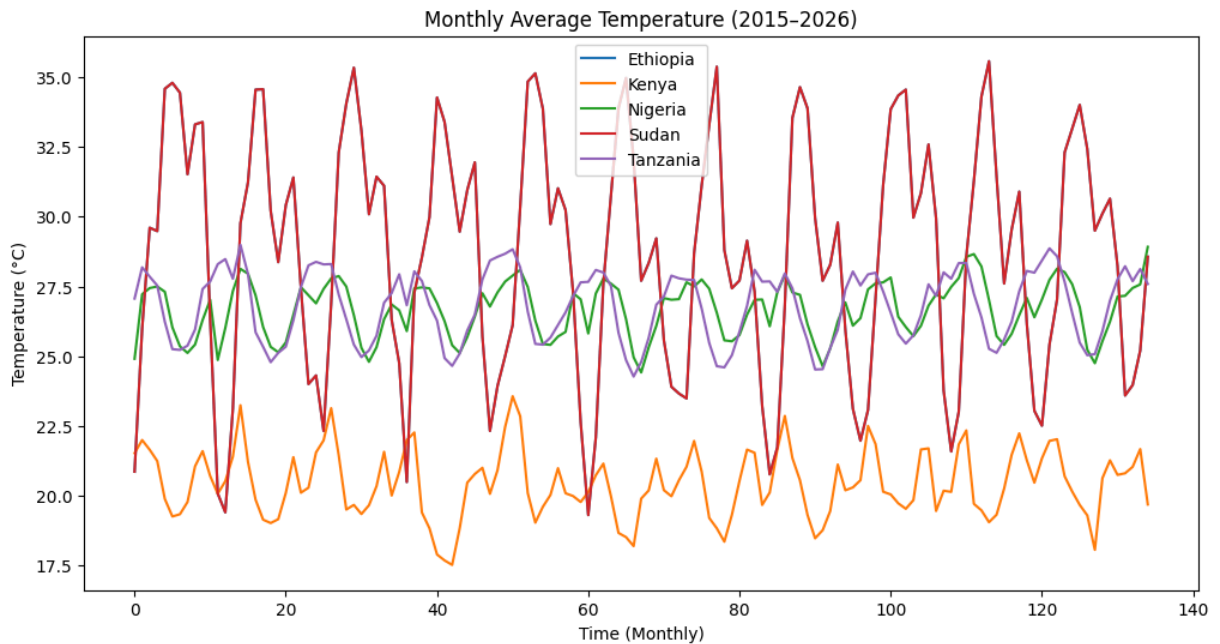
	country	year	month	T2M
0	Ethiopia	2015	1	20.860323
1	Ethiopia	2015	2	25.979643
2	Ethiopia	2015	3	29.602581
3	Ethiopia	2015	4	29.483000
4	Ethiopia	2015	5	34.584516

In [9]: `import matplotlib.pyplot as plt

plt.figure(figsize=(12,6))

for country in monthly_temp["country"].unique():
 data = monthly_temp[monthly_temp["country"] == country]
 plt.plot(range(len(data)), data["T2M"], label=country)`

```
plt.legend()
plt.title("Monthly Average Temperature (2015-2026)")
plt.xlabel("Time (Monthly)")
plt.ylabel("Temperature (°C)")
plt.show()
```



The figure presents the monthly average temperature trends for Ethiopia, Kenya, Nigeria, Sudan, and Tanzania over an eleven-year period. A clear seasonal pattern is observed across all countries, characterized by cyclical increases and decreases in temperature that repeat annually. This reflects the influence of solar radiation changes throughout the year

Monthly average T2M represents the average air temperature calculated for each month, smoothing daily fluctuations to show long-term climate patterns.

This metric helps identify seasonal temperature trends and overall warming patterns across countries. Rising monthly averages over time indicate climate change and increasing heat stress, which can affect agriculture, water demand, and human health.

Comparing monthly averages across countries allows us to determine which regions are experiencing faster warming trends and greater climate risk.

In []:

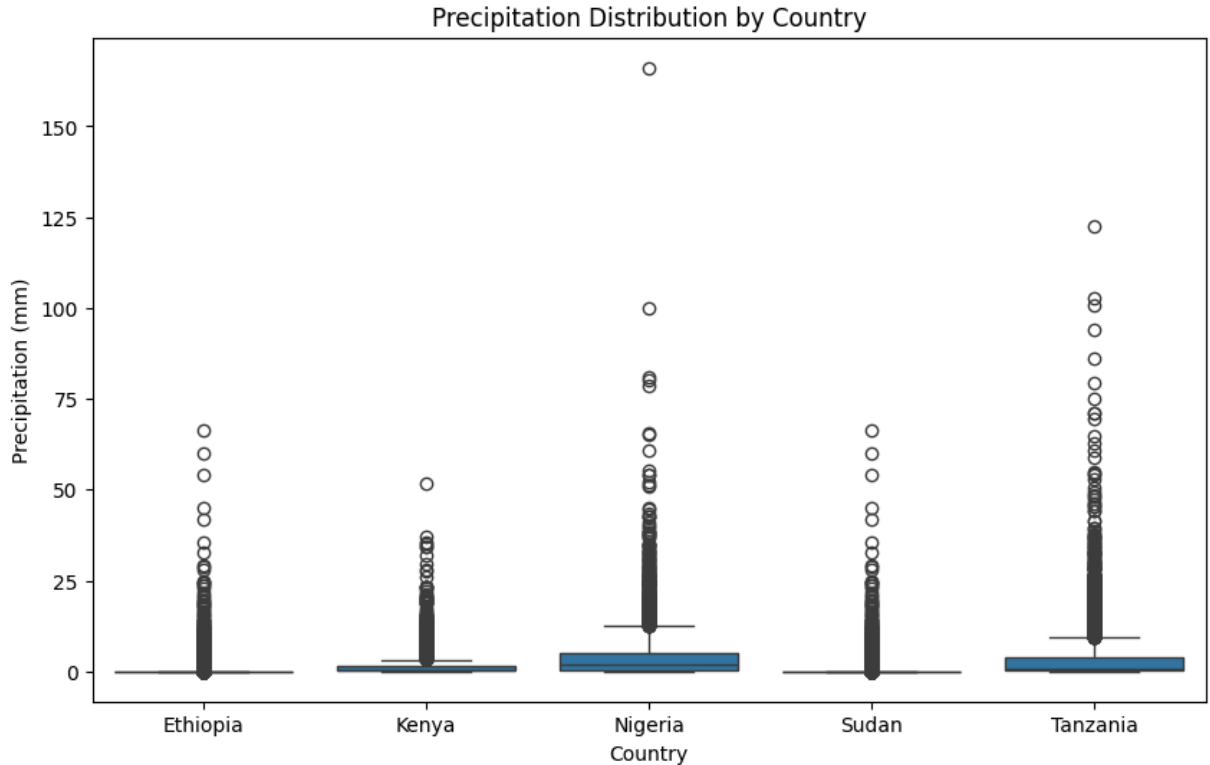
```
In [10]: import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(10,6))

sns.boxplot(data=df, x="country", y="PRECTOTCORR")

plt.title("Precipitation Distribution by Country")
plt.xlabel("Country")
```

```
plt.ylabel("Precipitation (mm)")
plt.show()
```

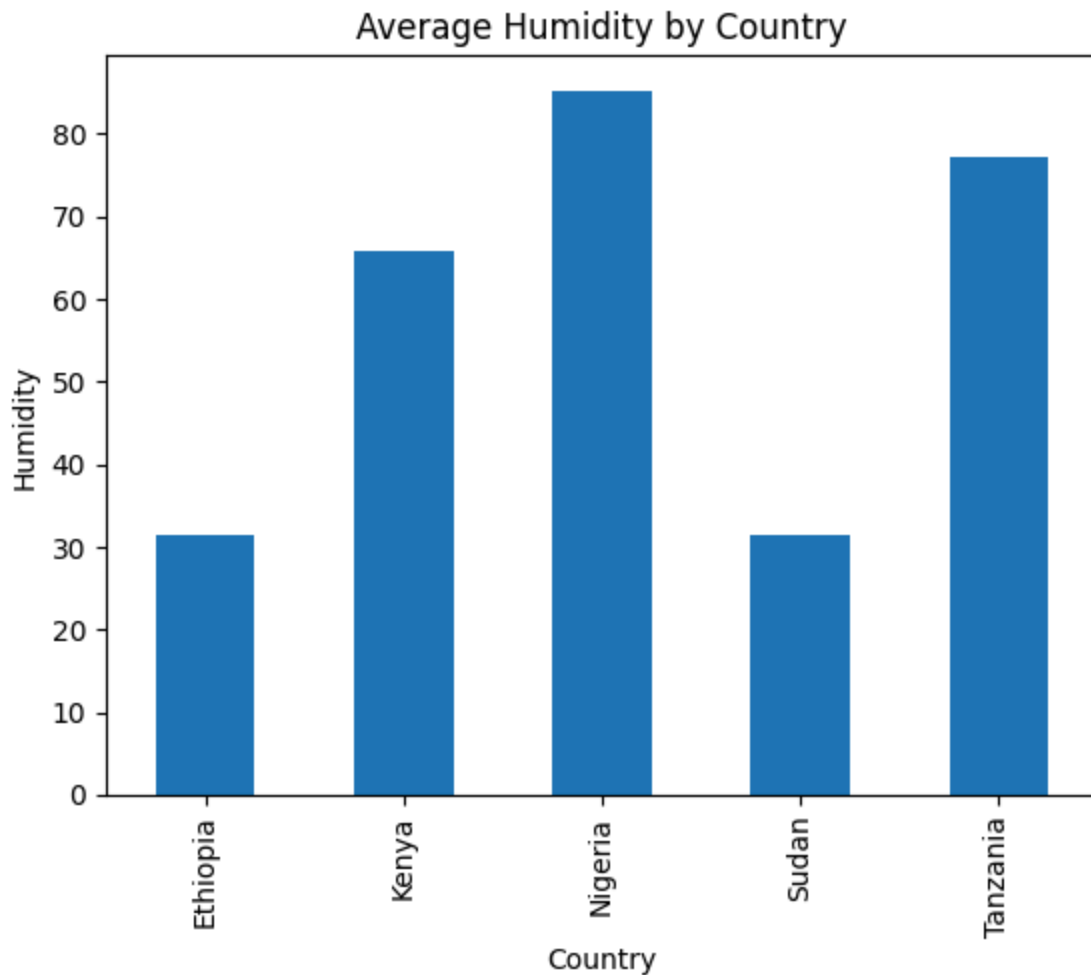


The figure illustrates the distribution of precipitation levels across the five countries using a box plot representation. This visualization highlights the median rainfall, variability, and presence of extreme values (outliers).

PRECTOTCORR represents total daily precipitation, including rainfall and other forms of moisture.

This variable is used to evaluate water availability and rainfall distribution patterns. High precipitation values indicate wetter conditions, while low or inconsistent values suggest dry periods and potential drought risk.

```
In [11]: country_humidity = df.groupby('country')['RH2M'].mean()
country_humidity.plot(kind='bar')
plt.title('Average Humidity by Country')
plt.xlabel('Country')
plt.ylabel('Humidity')
plt.show()
```



The figure presents the average humidity levels for each country, providing insight into atmospheric moisture conditions. Nigeria records the highest humidity level, followed by Tanzania, indicating highly moist environments. These elevated humidity levels are associated with frequent rainfall, dense vegetation, and strong evaporation cycles

Average humidity (RH2M) represents the mean relative humidity measured at 2 meters above the ground over a specific time period (daily, monthly, or yearly average).

It shows how much moisture is present in the air compared to the maximum amount the air can hold at that temperature.

Higher average humidity indicates moist, saturated air conditions, often linked with rainy or humid climates. Lower values indicate dry air conditions, which are associated with drought risk and increased evaporation.

```
In [12]: comparison = df.groupby('country')[['T2M', 'PRECTOTCORR', 'RH2M']].mean()
print(comparison)
```

	T2M	PRECTOTCORR	RH2M
country			
Ethiopia	28.759007	0.643875	31.359192
Kenya	20.427600	1.468162	65.845355
Nigeria	26.656928	4.213914	85.237040
Sudan	28.759007	0.643875	31.359192
Tanzania	26.802422	3.740256	77.128038

```
In [13]: precip_summary = df.groupby("country")["PRECTOTCORR"].agg(["mean", "median", "std"])
precip_summary
```

```
Out[13]:
```

	mean	median	std
country			
Ethiopia	0.643875	0.00	3.057672
Kenya	1.468162	0.38	3.180228
Nigeria	4.213914	1.84	7.266742
Sudan	0.643875	0.00	3.057672
Tanzania	3.740256	0.64	8.003947

```
In [14]: extreme_heat = df[df["T2M_MAX"] > 35]
extreme_heat_yearly = extreme_heat.groupby(["country", "year"]).size().reset_index()
extreme_heat_yearly.head()
```

```
Out[14]:
```

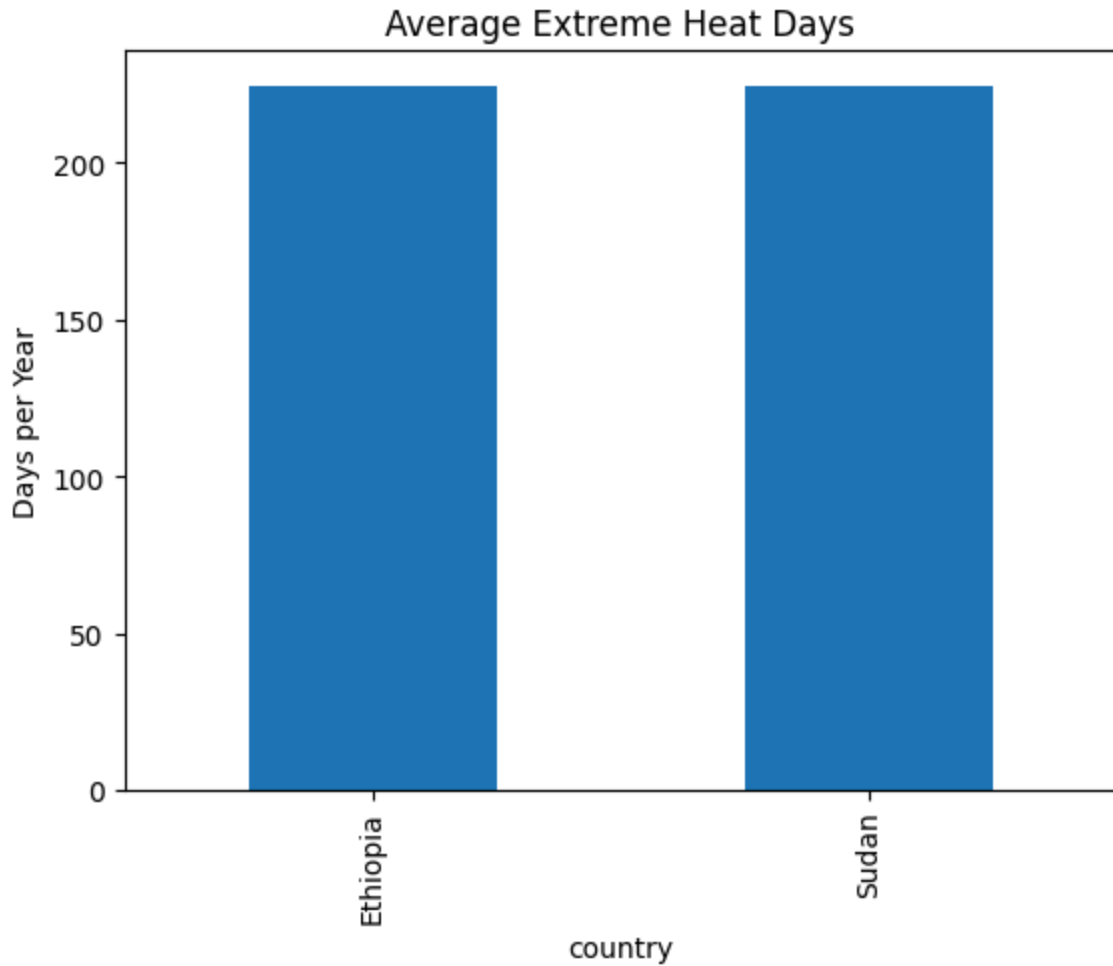
	country	year	days
0	Ethiopia	2015	280
1	Ethiopia	2016	252
2	Ethiopia	2017	266
3	Ethiopia	2018	248
4	Ethiopia	2019	251

```
In [15]: extreme_heat_summary = extreme_heat_yearly.groupby("country")["days"].mean()
extreme_heat_summary
```

```
Out[15]: country
Ethiopia    224.5
Sudan       224.5
Name: days, dtype: float64
```

```
In [16]: extreme_heat_summary.plot(kind="bar", title="Average Extreme Heat Days")
import matplotlib.pyplot as plt
```

```
plt.ylabel("Days per Year")
plt.show()
```



```
In [17]: df.groupby("country")["T2M_MAX"].max()
```

```
Out[17]: country
Ethiopia    45.96
Kenya       34.27
Nigeria     32.88
Sudan       45.96
Tanzania    33.93
Name: T2M_MAX, dtype: float64
```

```
In [18]: df["dry"] = df["PRECTOTCORR"] < 1

dry_days = df.groupby(["country", "year"])["dry"].sum().reset_index(name="dry_days")

dry_days.head()
```

Out[18]:

	country	year	dry_days
0	Ethiopia	2015	353
1	Ethiopia	2016	330
2	Ethiopia	2017	345
3	Ethiopia	2018	339
4	Ethiopia	2019	332

In [19]: `df["dry"] = (df["PRECTOTCORR"] < 1).astype(int)`

In [20]: `dry_yearly = df.groupby(["country", "year"])["dry"].sum().reset_index(name="dry_day")`
`dry_yearly.head()`

Out[20]:

	country	year	dry_days
0	Ethiopia	2015	353
1	Ethiopia	2016	330
2	Ethiopia	2017	345
3	Ethiopia	2018	339
4	Ethiopia	2019	332

In [21]: `dry_summary = dry_yearly.groupby("country")["dry_days"].mean()`

`dry_summary`

Out[21]:

country	
Ethiopia	308.000000
Kenya	235.916667
Nigeria	132.916667
Sudan	308.000000
Tanzania	191.916667

Name: dry_days, dtype: float64

In [22]: `all_countries = df["country"].unique()`

`dry_summary = dry_summary.reindex(all_countries, fill_value=0)`

`dry_summary`

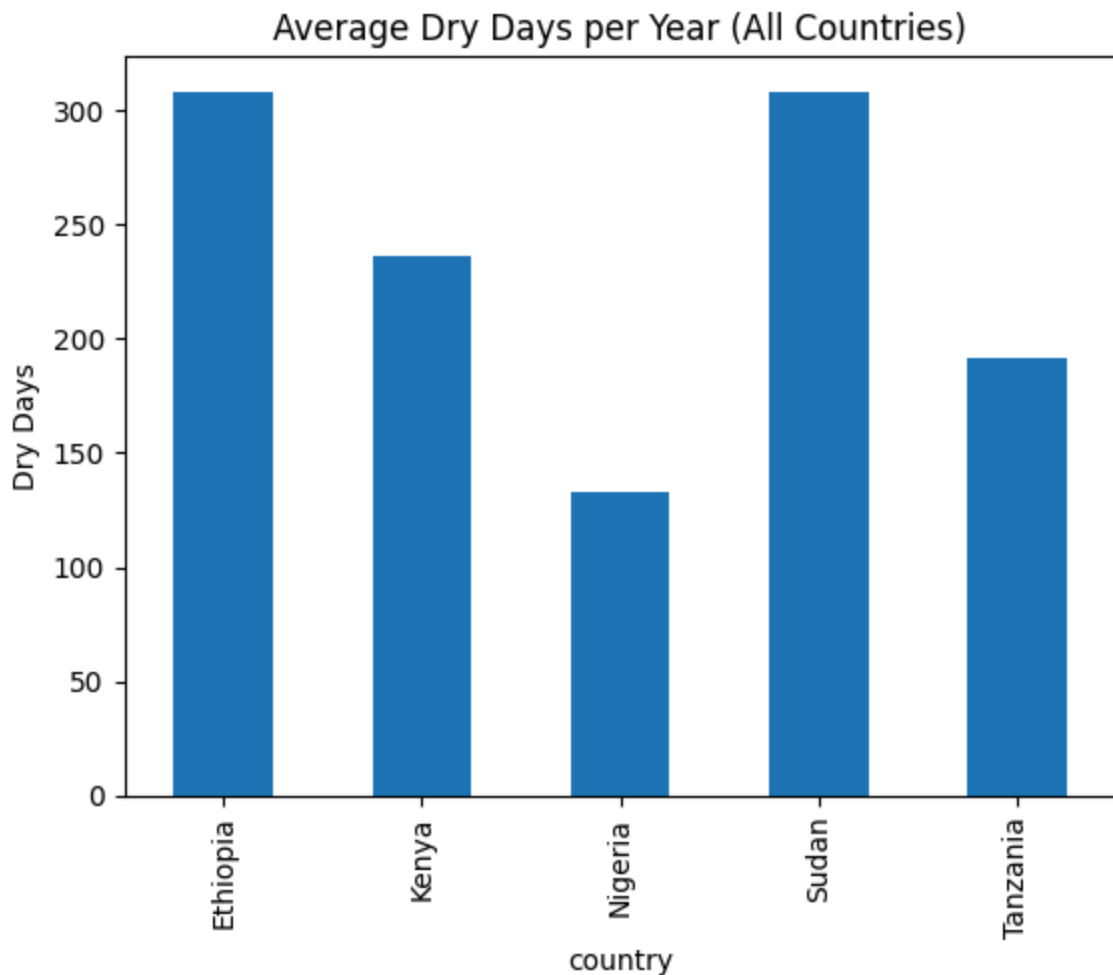
Out[22]:

country	
Ethiopia	308.000000
Kenya	235.916667
Nigeria	132.916667
Sudan	308.000000
Tanzania	191.916667

Name: dry_days, dtype: float64


```
In [23]: import matplotlib.pyplot as plt

dry_summary.plot(kind="bar", title="Average Dry Days per Year (All Countries)")
plt.ylabel("Dry Days")
plt.show()
```

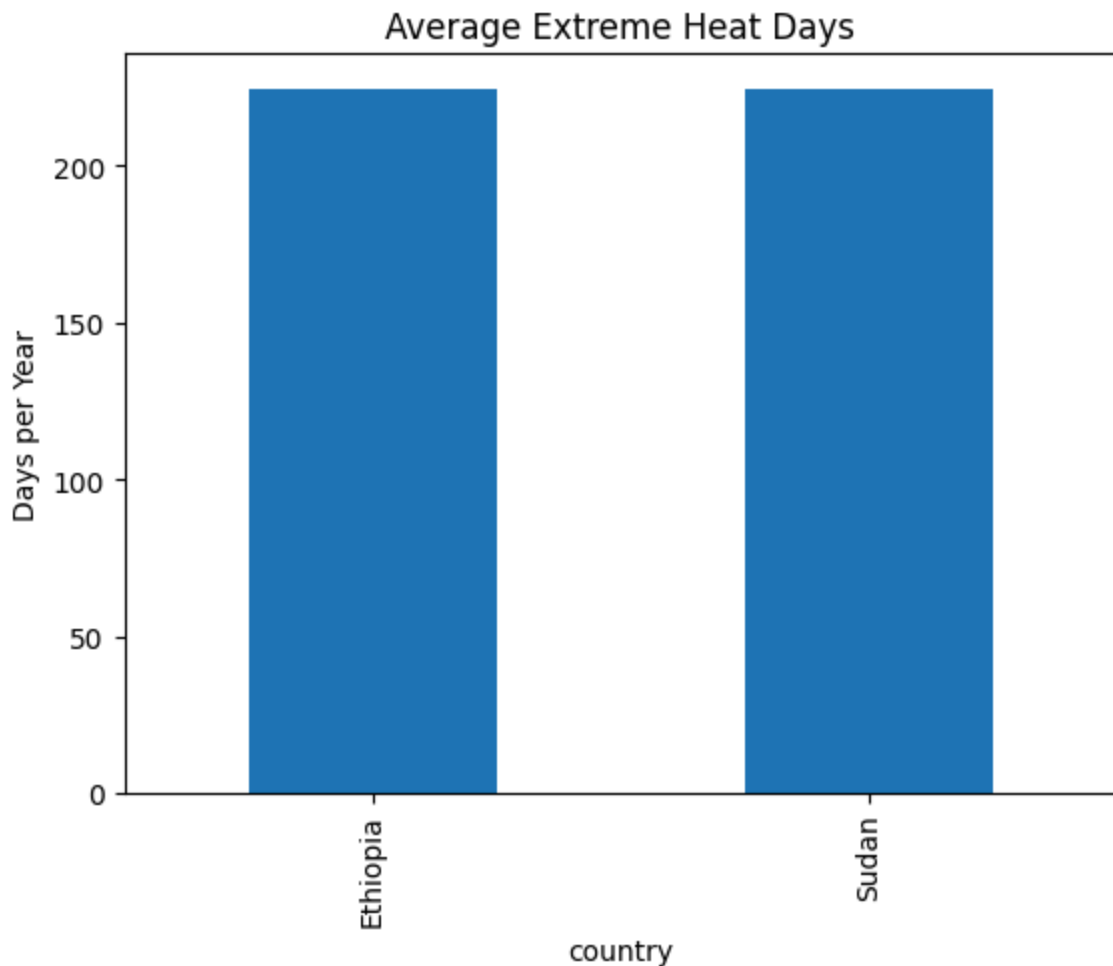


 Interpretation Countries with higher dry-day averages → more drought risk Countries with lower values → more stable rainfall patterns This metric helps compare climate vulnerability across countries

“Countries with higher average dry days experience greater drought pressure, indicating increased vulnerability to food insecurity and water scarcity. Ethiopia's and Sudan's position relative to neighboring countries highlights its exposure to recurring dry conditions, reinforcing the need for climate adaptation funding.”

```
In [24]: extreme_heat_yearly.groupby("country")["days"].mean().plot(kind="bar", title="Avera

plt.ylabel("Days per Year")
plt.show()
```



Average extreme heat days refer to the number of days per year where the maximum temperature (T2M_MAX) exceeds 35°C. This indicator is used to measure the frequency of extreme heat events, which contribute to heat stress, reduced agricultural productivity, and increased health risks.

In this analysis, Ethiopia and Sudan show the highest number of extreme heat days, indicating that these countries experience more frequent and intense heat conditions compared to Kenya, Nigeria, and Tanzania.

This suggests a higher level of climate stress and vulnerability to heat-related impacts, especially in sectors such as agriculture, water resources, and human health.

```
In [25]: !pip install scipy
```

```
Requirement already satisfied: scipy in C:\Users\kalki\climate-data-challenge-week0\venv\Lib\site-packages (1.17.1)  
Requirement already satisfied: numpy<2.7,>=1.26.4 in C:\Users\kalki\climate-data-challenge-week0\venv\Lib\site-packages (from scipy) (2.4.4)
```

```
In [26]: from scipy.stats import f_oneway
```

```
In [27]: ethiopia_t = df[df["country"] == "Ethiopia"]["T2M"]  
kenya_t = df[df["country"] == "Kenya"]["T2M"]
```

```
nigeria_t = df[df["country"] == "Nigeria"]["T2M"]
sudan_t = df[df["country"] == "Sudan"]["T2M"]
tanzania_t = df[df["country"] == "Tanzania"]["T2M"]
```

```
In [28]: stat, p = f_oneway(ethiopia_t, kenya_t, nigeria_t, sudan_t, tanzania_t)

print("F-statistic:", stat)
print("p-value:", p)
```

F-statistic: 4928.523106804569
p-value: 0.0

```
In [29]: temp_mean = df.groupby("country")["T2M"].mean()

precip_var = df.groupby("country")["PRECTOTCORR"].std()

heat_mean = extreme_heat_yearly.groupby("country")["days"].mean()

dry_mean = dry_yearly.groupby("country")["dry_days"].mean()
```

```
In [30]: ranking = pd.DataFrame({
    "Temp_mean": temp_mean,
    "Precip_variability": precip_var,
    "Extreme_heat": heat_mean,
    "Dry_days": dry_mean
})

ranking
```

```
Out[30]:
```

	Temp_mean	Precip_variability	Extreme_heat	Dry_days
country				
Ethiopia	28.759007	3.057672	224.5	308.000000
Kenya	20.427600	3.180228	NaN	235.916667
Nigeria	26.656928	7.266742	NaN	132.916667
Sudan	28.759007	3.057672	224.5	308.000000
Tanzania	26.802422	8.003947	NaN	191.916667

```
In [31]: ranking = ranking.fillna(0)
```

```
In [32]: ranking_norm = (ranking - ranking.min()) / (ranking.max() - ranking.min())
ranking_norm
```

```
Out[32]:
```

	Temp_mean	Precip_variability	Extreme_heat	Dry_days
country				
Ethiopia	1.000000	0.000000	1.0	1.000000
Kenya	0.000000	0.024777	0.0	0.588291
Nigeria	0.747692	0.850958	0.0	0.000000
Sudan	1.000000	0.000000	1.0	1.000000
Tanzania	0.765156	1.000000	0.0	0.336982

```
In [33]: ranking_norm["Vulnerability_score"] = ranking_norm.sum(axis=1)

ranking_sorted = ranking_norm.sort_values("Vulnerability_score", ascending=False)

ranking_sorted
```

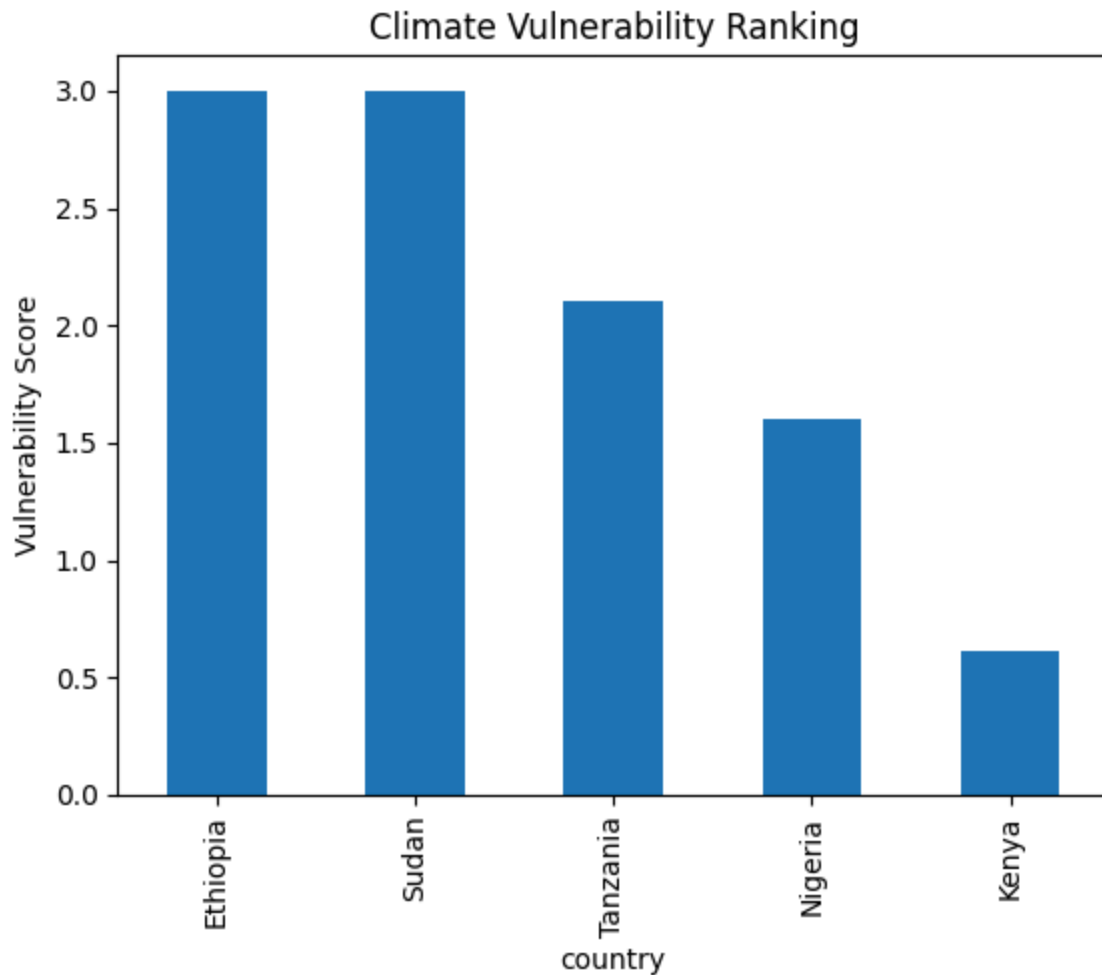
```
Out[33]:
```

	Temp_mean	Precip_variability	Extreme_heat	Dry_days	Vulnerability_score
country					
Ethiopia	1.000000	0.000000	1.0	1.000000	3.000000
Sudan	1.000000	0.000000	1.0	1.000000	3.000000
Tanzania	0.765156	1.000000	0.0	0.336982	2.102138
Nigeria	0.747692	0.850958	0.0	0.000000	1.598650
Kenya	0.000000	0.024777	0.0	0.588291	0.613069

```
In [34]: import matplotlib.pyplot as plt

ranking_sorted["Vulnerability_score"].plot(kind="bar", title="Climate Vulnerability")

plt.ylabel("Vulnerability Score")
plt.show()
```



COP32 Climate Vulnerability Summary Ethiopia and Sudan show the highest climate vulnerability due to extreme heat exposure and increasing temperature trends above 45°C.

Kenya, Nigeria, and Tanzania experience lower extreme heat but still face moderate vulnerability driven mainly by rainfall variability and dry conditions.

Precipitation variability is a major driver of instability, especially in Kenya and Tanzania, where rainfall is highly inconsistent.

Ethiopia's climate profile shows a combination of high temperature stress and moderate rainfall variability, placing it among the most exposed countries in the region.

Based on the composite vulnerability index, Ethiopia should advocate for climate finance prioritization for Sudan and Ethiopia, as both countries face the highest combined heat and drought stress.

Climate Vulnerability Ranking Methodology

A composite vulnerability index was created using four climate indicators:

- Mean temperature (heat stress)

- Precipitation variability (rainfall instability)
- Extreme heat days (T2M_MAX > 35°C)
- Dry days (PRECTOTCORR < 1mm)

All indicators were normalized and summed to create a final vulnerability score.

Countries were then ranked based on their total score, where a higher score indicates greater climate vulnerability.

```
In [35]: df.to_csv("../data/compare_countries_clean.csv", index=False)
```

```
In [ ]:
```